Exercise 3.6

1.(i) $(x+6)(x+4)$

(ii) $(z+6)(z-2)$

(iii) $(p-8)(p+2)$

(iv) $(t-9)(t-8)$

(v) $(y-20)(y+4)$

(vi) $(a+30)(a-20)$

2. (i) $(2a+5)(a+2)$

(ii) $(x-7y)(5x+6y)$ (iii) $(2x-3)(4x-3)$ (iv) $2(3x+2y)(x+2y)$

(v) $3x^2(3y+2)^2$ (vi) $(a+b+6)(a+b+3)$

3. (i) $(p-q-8)(p-q+2)$

(ii) $(m+6n)(m-4n)$ (iii) $\left(a+\sqrt{5}\right)\left(\sqrt{5}a-3\right)$ (iv) $(a+1)(a-1)(a^2-2)$

(v) $m(4m+5n)(2m-3n)$

(vi) $\left(\frac{1}{x}+\frac{1}{y}\right)^2$

Exercise 3.7

1. (i) Quotient : $4x^2-6x-5$, Remainder : 33 (ii) Quotient : $4y^2-6y+5$, Remainder : -10

(iii) Quotient : $4x^2+2x+1$, Remainder : 0 (iv) Quotient : $8z^2-6z+2$, Remainder : 10

2. Length : $x+4$ 3. Height : $5x-4$ 4. Mean : $x^2-5x+25$

5. (i) x^2+4x+5, 12 (ii) (x^2-1), -2

(iii) $3x^2-11x+40$, -125 (iv) $2x^3-\frac{x^2}{2}-\frac{3x}{8}+\frac{51}{32}$, $\frac{109}{32}$

6. $4x^3-2x^2+3$, $p=-2$, $q=0$, remainder=-10

7. $a=20$, $b=94$ & remainder=388

Exercise 3.8

1.(i) $(x-2)(x+3)(x-4)$ (ii) $(x+1)(x-2)(2x-1)$

(iii) $(x-1)(2x-1)(2x+3)$ (iv) $(x+2)(x+3)(x-4)$

(v) $(x-1)(x-2)(x+3)$ (vi) $(x-1)(x-10)(x+1)$

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